



CANDIDATE
NAME

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CENTRE
NUMBER

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CANDIDATE
NUMBER

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0478/22

May/June 2022

1 hour 45 minutes

No additional materials are needed.

- Answer **all** questions.
- **Do not attempt Tasks 1, 2 and 3** in the copy of the pre-release material on page 2; these are for information only.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- Do **not** write on any bar codes.
- Calculators must **not** be used in this paper.

- The total mark for this paper is 50.
- The number of marks for each question or part question is shown in brackets [].
- No marks will be awarded for using brand names of software packages or hardware.

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[Turn over

Section A

You are advised to spend no longer than 40 minutes answering this section.

Here is a copy of the pre-release material.

DO NOT attempt Tasks 1, 2 and 3 now.

Use the pre-release material and your experience from attempting the following tasks before the examination to answer Question 1.

Pre-release material

A program is needed to allow a Wildlife Park to sell tickets. A booking consists of one or more tickets for the same day(s) and can be made up to a week in advance. A booking can be made for a visit of one day or two consecutive days. A booking can have extra attractions included. A booking will be valid for the day(s) chosen only.

Ticket type	Cost for one day	Cost for two days
one adult	\$20.00	\$30.00
one child (an adult may bring up to two children)	\$12.00	\$18.00
one senior	\$16.00	\$24.00
family ticket (up to two adults or seniors, and three children)	\$60.00	\$90.00
groups of six people or more, price per person	\$15.00	\$22.50

Extra attraction	Cost per person
lion feeding	\$2.50
penguin feeding	\$2.00
evening barbecue (two-day tickets only)	\$5.00

Write and test a program or programs for the Wildlife Park:

- Your program or programs must include appropriate prompts for the entry of data. Data must be validated on entry.
- All outputs, including error messages, need to be set out clearly and understandably.
- All variables, constants and other identifiers must have meaningful names.

You will need to complete these **three** tasks. Each task must be fully tested.

Task 1 – displaying the ticket options and the extra attractions available

Set up your program to:

- display the options, attractions and prices for one-day tickets
- display the options, attractions and prices for two-day tickets
- show the days available for booking; assume that there are tickets available for any valid day.

Task 2 – process a booking

Extend your program for **Task 1** to:

- input the tickets and extra attractions required, then calculate the total cost of the booking
- allocate a unique booking number
- display the booking details, including the total cost and the unique booking number
- repeat as required.

Task 3 – ensuring each booking is the best value

Check that the total for each booking gives the best value and offer an alternative if this is **not** the case. For example, buying two family tickets is better than a group ticket for a group of 10 that includes four adults and six children.

1 All variables, constants and other identifiers must have meaningful names.

- (a) (i) Identify **one** constant that you could have used for **Task 1**.
Give the value and use of the constant.

Constant

Value

Use

[3]

- (ii) Identify **one** variable that you could have used for **Task 2**.
Give the data type and use of the variable.

Variable

Data type

Use

[3]

- (b) Explain how your program showed the days available for booking in **Task 1**.

.....

 [3]

- (c) Explain how your program made sure that each booking number allocated in **Task 2** was unique.

.....

 [2]

- (d) Write an algorithm for the part of **Task 2** that inputs the tickets and extra attractions required then calculates the total cost of the booking.

Assume that the booking is for a single day.

Use pseudocode, programming statements or a flowchart.

[illegible]

- (e) Explain how your program decides when a family ticket is better value in **Task 3**.

Any programming statements that you include in your answer must be fully explained.

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..... [3]

Section B starts on page 8

Section B

- 2 An algorithm allows a user to input their password and checks that there are at least eight characters in the password. Then, the user is asked to re-input the password to check that both inputs are the same. The user is allowed three attempts at inputting a password of the correct length and a matching pair of passwords. The pre-defined function `LEN(X)` returns the number of characters in the string, `X`

```

01  Attempt ← 0
02  REPEAT
03      PassCheck ← TRUE
04      OUTPUT "Please enter your password "
05      INPUT Password
06      IF LEN(Password) < 8
07          THEN
08              PassCheck ← TRUE
09          ELSE
10              OUTPUT "Please re-enter your password "
11              INPUT Password2
12              IF Password <> Password2
13                  THEN
14                      PassCheck ← FALSE
15              ENDIF
16          ENDIF
17      Attempt ← Attempt + 1
18  UNTIL PassCheck OR Attempt <> 3
19  IF PassCheck
20      THEN
21          OUTPUT "Password success"
22      ELSE
23          OUTPUT "Password fail"
24  ENDIF

```

- (a) Identify the **three** errors in the pseudocode and suggest a correction to remove each error.

Error 1

Correction

Error 2

Correction

Error 3

Correction

[3]

- (b) The algorithm includes **two** types of check on the data input. Identify and describe each type of check.

Type of check 1

Description

.....

Type of check 2

Description

.....

[4]

- (c) Give **two** sets of test data for this algorithm and a reason for choosing each set.

Each set of test data and its reason must be different.

Set 1

Reason

.....

Set 2

Reason

.....

[4]

- 3 (a) Describe a one-dimensional array. Include an example of an array declaration.

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.....

.....

.....

.....

..... [3]

- (b) Explain how indexing could be used to search for a value stored in a one-dimensional array.

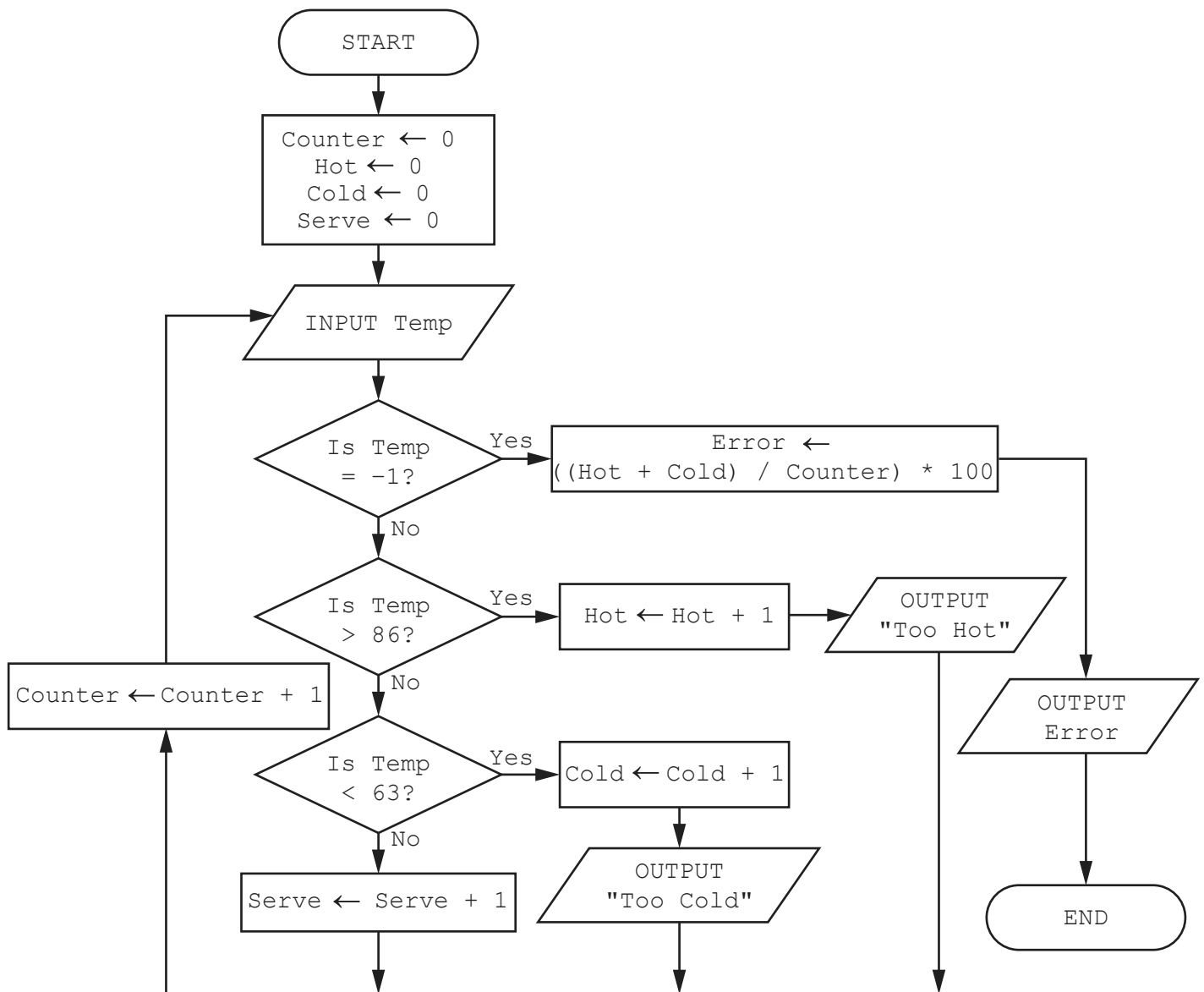
.....

.....

.....

..... [2]

- 4 This algorithm checks the temperature of hot food being served to customers.



(a) Complete the trace table for the algorithm using this input data:

75, 78, 84, 87, 91, 80, 75, 70, 65, 62, -1, 20

Counter	Hot	Cold	Serve	Temp	Error	OUTPUT

[7]

(b) State how the final output from the algorithm could be improved.

.....
 [1]

(c) Identify the process in the algorithm that is **not** required.

.....
 [1]

- 5 A database table, NURSE, is used to keep a record of disposable items worn by veterinary nurses.

This is part of the table:

ItemNumber	Description	SingleUse	Uses	StockLevel	ReorderLevel
DIG1	Glove (pair)	Y	1	500	800
DIA1	Apron	Y	1	700	800
DIM5	Hair net	Y	1	650	500
DIA2	Apron	N	5	25	100
DIS4	Suit	N	3	70	50
DIV9	Shoe cover (pair)	Y	1	400	250

- (a) Complete this query-by-example grid to display only the item number and the description of single use items, where the stock level is below the reorder level.

Field:					
Table:					
Sort:					
Show:	☐	☐	☐	☐	☐
Criteria:					
or:					

[4]

- (b) Give a reason why the field SingleUse is **not** required in the table NURSE.

.....

..... [1]

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